Failure Geometry: A Multi-Dimensional and Cross-Dataset Analysis of Object Detection Robustness under Real-World Corruptions

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***Abstract*—Object detection models routinely achieve strong results on standard benchmarks, yet their behaviour under real-world corruptions is much less well understood. Scalar metrics like mean Average Precision collapse all failure information into a single number — making it impossible to distinguish whether a model missed detections, generated false positives, lost localisation precision, or became overconfident.**

We introduce *Failure Geometry*, an evaluation framework that represents corruption-induced degradation as a four-dimensional failure vector $\mathbf{F}=[\Delta R, \Delta P, \Delta L, \Delta E]$, capturing recall degradation, precision shift, localisation error, and calibration drift. We evaluate four pretrained detectors — YOLOv8n, YOLOv9c, DETR (ResNet-50), and RT-DETR — under 17 corruption types across five severity levels on *two* independent benchmarks: the full COCO val2017 dataset (5,000 images) and Pascal VOC, substantially expanding prior work which covered at most nine corruption types on a single dataset and subsets of 500 to 2,000 images.

Our experiments surface two architecturally consistent failure modes that replicate across both datasets. *Hallucination* under zoom blur — where false-positive rate increases with severity — is confirmed on COCO for all four models (ΔP=$+0.264$ for YOLOv9c down to $+0.084$ for RT-DETR at severity 5) and, with attenuated magnitude, on three of four models on VOC (YOLOv8n $+0.068$, DETR $+0.137$, YOLOv9c ≈ 0.001), establishing hallucination as a largely corruption-intrinsic phenomenon while also revealing that its strength is dataset-dependent. *Threshold- collapse* under pixelation — near-maximal degradation already at severity 1 — and the newly identified *Late-Onset Collapse* under JPEG compression are both reproduced on VOC across all four architectures, confirming these as robust, dataset-independent failure signatures.

Beyond replicating the taxonomy qualitatively, we introduce three quantitative instruments that turn "does the taxonomy transfer?" into a measurable question. A cross-dataset transfer-prediction analysis (Ridge regression under leave-one-corruption-out cross-validation) shows that ΔR and ΔL transfer strongly from COCO to VOC (R^2=0.84–0.95 across three architectures), whereas ΔP and ΔE transfer weakly and unevenly (R^2 as low as 0.15 for DETR), explaining *why* threshold-collapse and late-onset-collapse — both ΔR-driven — replicate reliably while hallucination does not. We further introduce the Failure Geometry Transferability Score (FGTS), which quantifies per-architecture taxonomy stability directly (YOLOv8n $=0.706$, YOLOv9c $=0.647$, DETR $=0.471$), numerically confirming that DETR is the least dataset-stable architecture in our study. Finally, a two-way ANOVA on ΔP confirms a statistically significant model$\times$dataset interaction (F=8.41, p<0.001) even when restricted to the three architectures with complete COCO and VOC coverage, establishing that dataset-dependent hallucination behaviour is a general architecture-dependent phenomenon rather than an idiosyncrasy of a single model.

Cross-architecture ranking via the proposed Mean Corruption Robustness Score (mCRS) places RT-DETR highest on both datasets (mCRS$_{\text{COCO}}$=0.9304, mCRS$_{\text{VOC}}$=0.8916), but the ranking of the remaining three models is *not* dataset-invariant: DETR and YOLOv8n are statistically tied on COCO, whereas on VOC DETR falls to last place (mCRS=0.8371) while YOLOv9c and YOLOv8n remain close to their COCO-relative positions. We show this rank instability is attributable in part to a precision-denominator effect that inflates ΔP magnitude on the lower-object-density VOC images, and we discuss its implications for using single-dataset robustness rankings as deployment guidance.

***Index Terms*—Object detection, robustness, corruption, failure analysis, YOLOv8, YOLOv9c, DETR, RT-DETR, calibration, ImageNet-C, Pascal VOC, failure geometry, mCRS, cross-dataset generalisation, transferability**

I. Introduction

Deployed object detectors operate in environments that look nothing like the clean, well-lit, high-resolution images on which they are evaluated. Sensors introduce noise. Video codecs compress frames. Rain, fog, and motion blur distort spatial structure. These corruptions are not edge cases — they are the ordinary conditions of real deployment in autonomous driving, industrial inspection, surveillance, and medical imaging [1], [20].

The standard response to this gap has been benchmarking: measure how much mAP drops under each corruption type and report the result [8]–[10]. This approach has produced valuable empirical evidence, most notably the finding by Liu *et al.* that robustness and clean accuracy are largely decoupled [10]. But a drop in mAP tells us very little about *why* a model failed or what a practitioner should do about it, and — as we show in this work — a robustness ranking measured on a single dataset can itself be misleading if it does not transfer to a second, independently curated benchmark.

We argue that failure analysis for object detection requires both a *multi-dimensional* and a *cross-dataset* perspective. A model can fail through at least four distinct mechanisms: (i) it misses detections that were previously found (*recall degradation*, ΔR); (ii) it generates spurious detections that did not previously exist (*precision inflation*, ΔP); (iii) its remaining detections drift away from ground-truth boxes (*localisation error*, ΔL); or (iv) its confidence scores stop reflecting detection accuracy (*calibration shift*, ΔE). Each mechanism has a different physical cause and demands a different remedy. Crucially, whether a given failure mode is a genuine property of the corruption — as opposed to an artefact of one particular dataset's object density, class distribution, or annotation style — can only be established by repeating the analysis on an independent benchmark, and, we argue, by *measuring* that transferability rather than only describing it qualitatively.

Prior work has not measured all four failure dimensions simultaneously, nor validated its conclusions on more than one dataset, nor quantified *how much* of a robustness taxonomy actually transfers. Calibration under corruption, in particular, has received almost no attention despite its safety implications: a model that suppresses detections is problematic, but a model that confidently hallucinates non-existent obstacles may be worse [15], [16].

This work

We introduce *Failure Geometry* — an evaluation framework that decomposes corruption-induced degradation into the four dimensions above and represents each corruption-severity condition as a point in the resulting four-dimensional failure space. We further extend the framework to a second, independent benchmark (Pascal VOC) to test whether the resulting failure taxonomy is a property of corruptions in general or an artefact of COCO specifically, and we go beyond qualitative replication by introducing quantitative instruments for measuring transferability itself. The key technical contributions are:

- A **formal failure mode taxonomy** with explicit decision rules for four modes: Suppression, Hallucination, Threshold-Collapse, and Gradual Degradation, plus a fifth mode — *Late-Onset Collapse* — observed under JPEG compression across all architectures and, as we newly show, across both datasets evaluated.
- **Large-scale empirical evaluation** of four detectors (YOLOv8n, YOLOv9c, DETR, RT-DETR) under 17 corruption types $\times$ 5 severity levels on the full COCO val2017 dataset (5,000 images), yielding 85 evaluation conditions per model.
- **Cross-dataset validation on Pascal VOC**: the same four pretrained detectors, the same 17 corruptions, and the same four-dimensional failure vector are re-evaluated on Pascal VOC, making this — to our knowledge — the first robustness study to test the *consistency of a failure taxonomy*, rather than only mAP degradation, across two independently curated detection benchmarks.
- **Cross-dataset failure-vector transfer prediction**: using Ridge regression under leave-one-corruption-out cross-validation, we show that a model's COCO failure vector predicts its VOC failure vector with high accuracy for ΔR and ΔL (R^2=0.84–0.95) but low and architecture-dependent accuracy for ΔP and ΔE — to our knowledge the first quantitative test of *which components* of a multi-dimensional failure vector transfer across detection benchmarks, and a direct explanation for why some failure modes (threshold-collapse, late-onset-collapse) replicate reliably while others (hallucination) do not.
- The **Failure Geometry Transferability Score (FGTS)**, a new formal metric that quantifies, per architecture, the fraction of corruptions whose assigned failure-mode label is identical on COCO and VOC. FGTS numerically confirms the qualitative finding that DETR is the least dataset-stable architecture in our study (FGTS=0.471, versus 0.706 for YOLOv8n).
- A **formal statistical validation** of the architecture $\times$ dataset interaction underlying the hallucination finding: a two-way ANOVA on ΔP (model $\times$ dataset $\times$ corruption family) shows a highly significant model$\times$dataset interaction (F=8.41, p<0.001), confirming that dataset-dependent hallucination behaviour is not an idiosyncrasy of a single architecture but a broader, statistically robust phenomenon, while family$\times$dataset and model$\times$family interactions are not significant.
- **Cross-architecture and cross-dataset validation** of the hallucination finding: zoom-blur-induced false-positive inflation is confirmed across CNN-based (YOLOv8n, YOLOv9c), pure transformer (DETR), and hybrid (RT-DETR) architectures on COCO, and is reproduced with reduced magnitude for three of four models on VOC — while also exposing a dataset-dependent sign flip for RT-DETR that we analyse in detail.
- The **Mean Corruption Robustness Score** (mCRS), a normalised multi-dimensional robustness scalar, together with a safety-weighted variant mCRS$_w$. We show that the cross-architecture *ranking* produced by mCRS is only partially stable across datasets: RT-DETR is the most robust architecture on both COCO and VOC, but the relative ordering of DETR and the YOLO models changes between the two benchmarks.
- A **failure prediction experiment** demonstrating that severity 1–3 failure vectors predict severity 4–5 behaviour with R^2=0.76 (YOLOv9c) on COCO and R^2=0.83 (mean Overall_R2, Ridge regression) on VOC, providing empirical grounding for the claim that failure geometry is structured rather than stochastic on both benchmarks.

Our framework is complementary to existing large-scale benchmarks: where Liu *et al.* [10] prioritise breadth (twelve models, sixteen corruptions), we prioritise diagnostic depth across four failure dimensions, replication across two datasets, and — new to this extended study — direct quantitative measurement of how much of that taxonomy actually transfers.

II. Related Work

A. Object Detection Architectures

The four models we evaluate span the main contemporary architectural families. YOLOv8n [3] and YOLOv9c [4] belong to the YOLO lineage of single-stage convolutional detectors — YOLOv9 introduces Programmable Gradient Information (PGI) and a Generalised ELAN backbone, achieving higher accuracy than YOLOv8 at comparable speed. DETR [6] replaced the conventional detection pipeline with a transformer encoder-decoder and bipartite matching, eliminating anchor boxes and NMS at the cost of slower training convergence. RT-DETR [5] retains the transformer decoder but replaces the CNN backbone with a hybrid feature extractor, recovering real-time speed while matching DETR-level accuracy. Two-stage anchor-based detectors such as Faster R-CNN [7] remain in wide deployment but are outside the scope of this study.

B. Robustness Benchmarks

Hendrycks and Dietterich [9] established the ImageNet-C protocol — fifteen synthetic corruption types at five severity levels — as the standard for measuring classifier robustness. Michaelis *et al.* [8] extended this to object detection, introducing COCO-C and demonstrating 30–60% mAP loss across standard detectors. Liu *et al.* [10] subsequently ran the most comprehensive study to date: twelve representative detectors, sixteen corruptions, two datasets (COCO-C and BDD100K-C). Their central conclusion — that a high clean-mAP detector can simultaneously be brittle under corruption — directly motivates our decomposition approach. Notably, even this two-dataset study reports only *aggregate mAP degradation* on each dataset; it does not ask whether a multi-dimensional failure taxonomy computed on one dataset survives evaluation on the other, nor does it attempt to *quantify* transferability with a predictive model or a dedicated metric. This is precisely the question our COCO/VOC comparison, transfer-prediction analysis, and FGTS metric are designed to answer.

Everingham *et al.* [2] introduced the Pascal VOC detection benchmark, which differs from COCO in object density (VOC images typically contain fewer, larger objects across 20 categories rather than 80), scene composition, and annotation protocol. These differences make VOC a natural stress test for whether a corruption-robustness taxonomy generalises or is an artefact of COCO's particular statistics.

Mao *et al.* [11] introduced COCO-O, focusing on natural distribution shifts (weather, sketch, cartoon) rather than synthetic corruptions; Chhipa *et al.* [12] extended this to open-vocabulary detectors. Neither work decomposes failure beyond mAP, and neither cross-validates its findings on a second detection dataset or proposes a transferability metric.

C. Calibration and Reliability

Guo *et al.* [14] showed that modern neural networks become systematically overconfident under distribution shift. Pathiraja *et al.* [15] extended this finding specifically to object detectors, demonstrating miscalibration in both in-domain and out-of-domain settings. Despite this, no existing robustness benchmark includes calibration as an evaluation dimension alongside detection performance. We close this gap by including ΔE (ECE shift) as the fourth dimension of our failure vector, on two datasets, and by showing that ΔE is itself among the weakest-transferring dimensions across datasets.

D. Failure Analysis and Texture Bias

Geirhos *et al.* [18] showed that ImageNet-trained CNNs are systematically biased toward high-frequency texture features rather than global shape, making them sensitive to noise corruptions that perturb local statistics. This bias partially explains our empirical observation that YOLOv8n and YOLOv9c show higher recall degradation than DETR under impulse noise on COCO. Adversarial robustness work [16], [17] has documented hallucination-like behaviour under structured perturbations; we show analogous effects arise under *natural* corruptions, and that their strength is itself modulated by the evaluation dataset — a modulation we now show is statistically significant at the model$\times$dataset level.

E. Research Gap

Table I summarises how this work relates to prior benchmarks. The key distinctions are dimensionality — we are the first to jointly report ΔR, ΔP, ΔL, and ΔE with a formal failure taxonomy — cross-dataset replication of that taxonomy, and, new to this extended study, quantitative measurement of that transferability via a predictive model, a dedicated transferability score, and a formal statistical interaction test, none of which any prior work in this line attempts.

III. Proposed Methodology

A. Problem Formulation

Let $\mathcal{M}$ be a pretrained object detector and I an image from a detection benchmark $\mathcal{D} \in \{\text{COCO val2017}, \text{Pascal VOC}\}$ [1], [2]. A corruption operator $\mathcal{C}(\cdot, s)$ at severity $s \in \{1, \dots, 5\}$ produces a degraded image:

$$I' = \mathcal{C}(I, s). \tag{1}$$

Running $\mathcal{M}$ on both I and I' yields clean predictions $\mathcal{M}(I)$ and corrupted predictions $\mathcal{M}(I')$. The per-image failure vector $\mathbf{F}$ measures degradation across four axes:

$$\mathbf{F} = [\Delta R,\ \Delta P,\ \Delta L,\ \Delta E]. \tag{2}$$

Every quantity below is computed identically on both datasets so that differences in the resulting failure geometry can be attributed to the data rather than to the metric definitions.

B. Four-Dimensional Failure Vector

Detections are matched to ground truth by greedy IoU assignment at threshold 0.5, consistent with the COCO protocol [1]. Let TP_c, FP_c, FN_c denote true positives, false positives, and false negatives under clean conditions, and similarly for the corrupted case.

TABLE I

COMPARISON OF FAILURE GEOMETRY WITH PRIOR OBJECT DETECTION ROBUSTNESS BENCHMARKS. ✓ = FULLY ADDRESSED; ○ = PARTIALLY ADDRESSED; × = NOT ADDRESSED.

Method / Work	Venue	Models	Datasets	Failure Dimensions				Taxonomy / mCRS	Quantified transferability
				ΔR	ΔP	ΔL	ΔE		
Michaelis *et al.* [8]	NeurIPS-W'19	6	3	○	×	×	×	×	×
Liu *et al.* [10]	IJCV'24	12	2	○	×	×	×	×	×
Mao *et al.* [11]	ICCV'23	20	1	○	×	×	×	×	×
Chhipa *et al.* [12]	ECCV-W'24	3	3	○	×	×	×	×	×
Hendrycks & Dietterich [9]	ICLR'19	–	ImageNet-C	○	×	×	×	×	×
Failure Geometry (Ours)	–	**4**	**2 (COCO, VOC)**	✓	✓	✓	✓	✓	✓

○ for ΔR in prior work indicates that recall-like information is partially recoverable from mAP, but is not independently reported or analysed. ΔP, ΔL, ΔE are not reported by any prior work. "Quantified transferability" refers to a predictive model, a dedicated transferability metric, or a formal interaction test of cross-dataset consistency; no prior work reports any of these.

$$\Delta R = \frac{TP_c - TP_d}{n_{GT}}, \qquad \Delta P = \frac{FP_d - FP_c}{n_{GT}} \tag{3}$$

where $n_{GT} = \max(1, |GT|)$ is the number of ground-truth boxes. A positive ΔR means objects are being missed; positive ΔP means false positives have *increased* — what we term hallucination. Because n_{GT} is the mean number of ground-truth boxes *per image*, and Pascal VOC images contain substantially fewer annotated objects on average than COCO images, the same absolute change in false positive count produces a larger ΔP magnitude on VOC. We return to this denominator effect in Section IV-C.

Localisation quality is measured via Generalised IoU [19]:

$$\Delta L = \overline{\text{GIoU}}_c - \overline{\text{GIoU}}_d \tag{4}$$

where the mean is taken over matched true-positive detections only. Calibration is measured by Expected Calibration Error over ten confidence bins [14]:

$$\Delta E = \text{ECE}_d - \text{ECE}_c. \tag{5}$$

C. Corruption Protocol

We apply 17 corruption types spanning four families at five severity levels each (85 evaluation conditions per model per dataset), extending the nine corruptions of the original ImageNet-C protocol [9]:

- **Noise (4):** Gaussian noise, shot noise, impulse noise, speckle noise.
- **Blur (4):** Defocus blur, motion blur, zoom blur, glass blur.
- **Weather (4):** Fog, frost, snow, spatter.
- **Digital (5):** JPEG compression, pixelation, brightness, contrast, elastic transform.

All corruptions are implemented in pure NumPy and OpenCV to avoid dependence on the `imagecorruptions` package, enabling exact reproducibility across platforms and datasets. Severity parameters are calibrated to match the visual intensity of the original ImageNet-C implementations and are held fixed between the COCO and VOC runs.

D. Failure Space

Per-image failure vectors are aggregated across the evaluation set to produce a mean vector $\bar{\mathbf{F}}_{c,s}$ for each corruption–severity pair, separately for each dataset. These $17 \times 5 = 85$ mean vectors form a *failure space* in $\mathbb{R}^4$ per dataset. Hierarchical clustering with Ward linkage on this space reveals whether corruption types form natural groupings, and linear regression across severity levels (R^2) quantifies trajectory predictability. Repeating this analysis on both COCO and VOC lets us test whether the resulting cluster topology and taxonomy assignments are dataset-specific or corruption-intrinsic.

E. Formal Failure Mode Taxonomy

Decision rules are applied to each corruption's trajectory to assign one or more failure modes, identically on both datasets:

Definition 1 (Suppression): $\Delta R > 0.3$ and $\Delta P < 0$ at severity 3 — dominant recall loss with *fewer* false positives.

Definition 2 (Hallucination): $\Delta P > 0$ at severity ≥ 3 — net false-positive inflation under corruption.

Definition 3 (Threshold-Collapse): $\text{Var}(\Delta R)_{s \in 1..5} < 0.01$ *and* $\max(\Delta R) > 0.15$ — severity-invariant catastrophic failure that saturates at severity 1.

Definition 4 (Gradual Degradation): $R^2(\Delta R) > 0.85$ across severities — linear, predictable recall trajectory.

Definition 5 (Late-Onset Collapse): $R^2(\Delta R) < 0.80$ *and* $\Delta R_{s=5} > 3 \times \Delta R_{s=3}$ — sub-linear growth that terminates in a sharp high-severity collapse. This mode was first identified on COCO and, as we show in Section IV-C, reproduces on VOC for JPEG compression across all four architectures.

A corruption may satisfy more than one definition simultaneously (e.g. a corruption can be both a Suppression and a Threshold-Collapse case); we record the full label *set* assigned to each corruption on each dataset, which is the input to the FGTS metric introduced below.

F. Mean Corruption Robustness Score

For each model and dataset, dimensions are normalised by their observed maximum across all conditions *within that* *dataset*:

$$\hat{F}_{c,s,d} = \frac{|\bar{F}_{c,s,d}|}{\max_{c',s'} |\bar{F}_{c',s',d}|}. \quad (6)$$

The mCRS is then:

$$\text{mCRS} = 1 - \frac{1}{|\mathcal{C}| \cdot |\mathcal{S}|} \sum_{c,s} \frac{\|\hat{\mathbf{F}}_{c,s}\|_2}{|\mathbf{F}|} \quad (7)$$

where $|\mathbf{F}|$=4. mCRS $\in [0, 1]$; higher values indicate greater robustness. Per-dataset normalisation means mCRS values are comparable *within* a dataset but should be read as relative rankings rather than absolute cross-dataset scores; we therefore report mCRS separately for COCO and VOC and discuss the ranking, not the raw magnitude, as the object of cross-dataset comparison. A safety-weighted variant mCRS$_w$ assigns weight 2 to ΔR and ΔL (reflecting their primacy in safety-critical deployment) and weight 1 to ΔP and ΔE, normalised so the total weight is 6.

G. Cross-Dataset Failure-Vector Transfer Prediction

To move from a qualitative claim of replication to a quantitative one, we test whether a model's COCO failure vector *predicts* its VOC failure vector at the level of individual corruption–severity conditions. For each model $\mathcal{M}$, let $\mathbf{x}_{c,s} = \bar{\mathbf{F}}^{\text{COCO}}_{c,s} \in \mathbb{R}^4$ and $\mathbf{y}_{c,s} = \bar{\mathbf{F}}^{\text{VOC}}_{c,s} \in \mathbb{R}^4$ be the paired COCO and VOC failure vectors for corruption c at severity s. We fit a multi-output Ridge regression $\mathbf{y} = W\mathbf{x} + b$ and evaluate it under *leave-one-corruption-out* cross-validation: for each of the 17 corruptions, all five of its severity levels are held out as the test fold, and the model is trained on the remaining 16 corruptions' 80 conditions. This grouping prevents leakage between severities of the same corruption, which would otherwise inflate R^2. We report both an overall R^2 (uniform average across the four output dimensions) and a per-dimension R^2, since our central question is precisely *which* components of the failure vector transfer and which do not.

H. Failure Geometry Transferability Score (FGTS)

The formal taxonomy of Section III-E assigns each corruption c, for a given model and dataset, a label set $L_{\mathcal{D}}(c) \subseteq$ {Suppression, Hallucination, Threshold-Collapse, Gradual, Late-Onset-Collapse, None} from Definitions 1–5. We define the Failure Geometry Transferability Score for a model $\mathcal{M}$ as the fraction of corruptions whose COCO and VOC label sets are identical:

$$\text{FGTS}(\mathcal{M}) = \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} \mathbb{1}[L_{\text{COCO}}(c) = L_{\text{VOC}}(c)]. \quad (8)$$

FGTS $\in [0, 1]$; a value of 1 means every corruption is assigned exactly the same failure mode(s) on both datasets, while 0 means no corruption's taxonomy label survives the transfer. Unlike the mCRS ranking-stability discussion in Section IV-C, which concerns the *relative ordering of models*, FGTS concerns the *consistency of a single model's own diagnostic labels* across datasets, and can be computed independently for any architecture with complete taxonomy data on both benchmarks.

I. Statistical Validation of the Architecture×Dataset Interaction

The hallucination finding (Section IV-C) is, on its face, driven by a single striking observation: RT-DETR's zoom-blur ΔP flips sign between COCO and VOC. To test whether dataset-dependent hallucination behaviour is a broader, statistically robust phenomenon rather than an artefact of one architecture, we fit a two-way ANOVA on severity-5 ΔP with factors Model, Dataset (COCO/VOC), and Corruption Family (Noise/Blur/Weather/Digital), restricted to the three architectures (YOLOv8n, YOLOv9c, DETR) with complete ΔP data on both benchmarks:

$$\Delta P \sim \text{Model}+\text{Dataset}+\text{Family}+\text{Model}\times\text{Dataset}+\text{Family}\times\text{Dataset}+\text{Mod} \quad (9)$$

A significant Model×Dataset interaction term would indicate that *how* ΔP shifts between datasets itself depends on which architecture is being evaluated — precisely the claim underlying our discussion of RT-DETR's sign flip, now tested formally and, crucially, without relying on RT-DETR's own (incomplete) COCO data.

IV. Experiments and Results

A. Experimental Setup

We evaluate four pretrained detectors, without any dataset-specific fine-tuning, on two benchmarks:

- **COCO val2017** [1] — 5,000 images, 36,781 ground-truth boxes, 80 categories.
- **Pascal VOC** [2] — 20 categories, evaluated with the same 17 corruptions × 5 severities protocol.

Model configurations are held fixed across both datasets:

- **YOLOv8n** [3] — CSPDarknet backbone, nano variant. Confidence threshold 0.25.
- **YOLOv9c** [4] — GELAN backbone with PGI, compact variant (≈51M parameters). Confidence threshold 0.25.
- **DETR** [6] — ResNet-50 encoder, transformer decoder. Confidence threshold 0.70.
- **RT-DETR** [5] — hybrid backbone with transformer decoder, large variant (≈67M parameters). Confidence threshold 0.25.

All models use the same pretrained (COCO-trained) weights on both evaluation sets; no VOC-specific fine-tuning is performed. This is a deliberate methodological choice: it allows us to isolate the effect of *evaluation-image statistics* (object density, scene composition, category distribution) on the failure geometry, holding the model itself constant. We discuss the resulting caveats in Section V-F. IoU threshold for TP/FP assignment is 0.5 throughout, on both datasets. Because RT-DETR's COCO evaluation is missing archived per-corruption vectors for several low-degradation conditions (Section V-F, limitation 5), the transfer-prediction (Section IV-C3), FGTS (Section IV-C4), and ANOVA (Section IV-C5) analyses below are computed on the three architectures with complete COCO and VOC coverage (YOLOv8n, YOLOv9c, DETR); mCRS rankings and the zoom-blur case study continue to include RT-DETR, for which complete VOC data is available.

TABLE II
COCO VAL2017: OVERALL AND PER-FAMILY MCRS.

Model	mCRS	mCRS$_w$	Noise	Blur	Weather	Digital
RT-DETR	**0.9304**	**0.9412**	**0.9523**	0.8819	0.9506	0.9355
YOLOv9c	0.9002	0.9182	0.9153	0.8239	**0.9903**	0.8769
DETR	0.8734	0.8959	0.8449	0.8041	0.9762	0.8693
YOLOv8n	0.8728	0.8956	0.844	**0.8034**	0.9758	**0.8687**

TABLE III
PASCAL VOC: OVERALL AND PER-FAMILY MCRS (SAME MODELS, CORRUPTIONS, AND METRIC DEFINITIONS AS TABLE II).

Model	mCRS	mCRS$_w$	Noise	Blur	Weather	Digital
RT-DETR	**0.8916**	**0.9202**	0.8882	0.8031	0.9734	**0.8995**
YOLOv9c	0.8907	0.9151	**0.8909**	**0.8247**	**0.9875**	0.8659
YOLOv8n	0.8694	0.8944	0.8290	0.8120	0.9741	0.8640
DETR	0.8371	0.8610	0.7668	0.7532	0.9676	0.8559

TABLE IV
OVERALL MCRS: COCO VS. VOC, SHOWING THE RANK SHIFT. RANK IN PARENTHESES.

Model	mCRS (COCO)	mCRS (VOC)
RT-DETR	0.9304 (1)	0.8916 (1)
YOLOv9c	0.9002 (2)	0.8907 (2)
DETR	0.8734 (3–4)	0.8371 (4)
YOLOv8n	0.8728 (3–4)	0.8694 (3)

TABLE V
ZOOM-BLUR ΔP AT SEVERITY 5: COCO VS. VOC.

Model	ΔP (COCO)	ΔP (VOC)
YOLOv9c	$+0.264$	$+0.001$
DETR	$+0.226$	$+0.137$
YOLOv8n	$+0.225$	$+0.068$
RT-DETR	$+0.084$	-0.762

B. COCO Results Summary

Table II reports the COCO mCRS ranking, reproduced from our original single-dataset study. RT-DETR achieves the highest overall robustness (mCRS=0.9304); DETR and YOLOv8n are statistically indistinguishable (=0.8734 vs. 0.8728).

C. Cross-Dataset Validation on Pascal VOC

We now repeat the identical evaluation pipeline on Pascal VOC to test whether the COCO-derived failure taxonomy and mCRS ranking generalise.

1) mCRS ranking on VOC: Table III reports the VOC mCRS ranking. RT-DETR remains the most robust architecture (mCRS=0.8916), consistent with COCO. However, the ordering of the remaining three models changes: YOLOv9c (=0.8907) and YOLOv8n (=0.8694) retain relatively strong robustness, while DETR falls to *last* place (mCRS=0.8371) — a materially larger gap than the near-tie observed between DETR and YOLOv8n on COCO. DETR's weather-family robustness remains high on VOC (=0.9676), but its noise (=0.7668) and blur (=0.7532) robustness are the lowest of any model on either dataset.

Table IV makes the rank shift explicit: RT-DETR and YOLOv9c hold their COCO ranks on VOC, but DETR and YOLOv8n swap places, with DETR's drop being the larger effect. This is the central cross-dataset finding of this work — *a single-dataset mCRS ranking should not be treated as a dataset-independent robustness ordering*, even though the identity of the *most* robust architecture (RT-DETR) does transfer. Sections IV-C3–IV-C5 below quantify *why* this instability occurs and how far it extends.

2) Hallucination under zoom blur, revisited: Table V compares zoom-blur-induced ΔP at severity 5 across both datasets. On COCO, all four models hallucinate (positive ΔP). On VOC, YOLOv8n and DETR still hallucinate, YOLOv9c is essentially neutral (≈ 0.001), and RT-DETR flips to a large *negative* value (-0.762).

We attribute this divergence to two compounding factors. First, the ΔP denominator n_{GT} is the mean number of ground-truth boxes per image; VOC images contain markedly fewer annotated objects than COCO images, so a given absolute change in false-positive count is amplified into a larger-magnitude ΔP. This is visible across *all* corruptions for RT-DETR on VOC, not only zoom blur — its VOC failure vectors show ΔP values in the -0.06 to -1.08 range (e.g. -1.075 under motion blur), an order of magnitude larger than the -0.01 to -0.28 range observed for the other three models on the same dataset. Second, RT-DETR's hybrid backbone and different NMS behaviour under VOC's larger, less cluttered objects may suppress the boundary- replica false positives that drive hallucination on COCO. Taken together, we conclude that *zoom-blur hallucination is a real and partially corruption-intrinsic phenomenon*, since it survives in three of four architectures on a second dataset, but that its magnitude and even its sign are sensitive to dataset-specific object statistics — a nuance invisible to any single-dataset study, and one we now show (Section IV-C5) is statistically systematic rather than anecdotal.

3) Cross-dataset failure-vector transfer prediction: Table VI reports leave-one-corruption-out R^2 for predicting each model's VOC failure vector from its COCO failure vector (Section III-G). Two of the four failure dimensions transfer strongly and consistently across all three architectures: ΔR (R^2=0.885–0.953) and ΔL (R^2=0.836–0.924). The remaining two dimensions transfer weakly and *unevenly*: ΔP ranges from 0.699–0.735 (YOLOv8n, YOLOv9c) down to just 0.151 for DETR, and ΔE is weak or slightly negative for all three models (as low as -0.068 for YOLOv9c, indicating COCO's calibration shift carries essentially no predictive information about VOC's calibration shift for that architecture).

This result directly explains the pattern of replication observed elsewhere in this study: threshold-collapse and late-onset-collapse are both defined purely in terms of the ΔR trajectory (Definitions 3 and 5), and ΔR is precisely the

TABLE VI
LEAVE-ONE-CORRUPTION-OUT R^2 FOR PREDICTING VOC FAILURE VECTORS FROM COCO FAILURE VECTORS (RIDGE REGRESSION).

Model	Overall	ΔR	ΔP	ΔL	ΔE
YOLOv8n	0.731	**0.953**	0.735	**0.924**	0.313
YOLOv9c	0.599	**0.930**	0.699	**0.836**	−0.068
DETR	0.606	**0.885**	0.151	**0.921**	0.466

TABLE VII
FAILURE GEOMETRY TRANSFERABILITY SCORE (FGTS) PER ARCHITECTURE.

Model	FGTS	Matching corruptions
YOLOv8n	**0.706**	12/17
YOLOv9c	0.647	11/17
DETR	0.471	8/17
Pooled	0.608	31/51

TABLE VIII
TWO-WAY ANOVA ON SEVERITY-5 ΔP (YOLOV8N, YOLOV9C, DETR; n=102 CORRUPTION×MODEL×DATASET OBSERVATIONS).

Factor	df	F	p
Model	2	7.06	0.0015
Dataset	1	18.12	0.00005
Family	3	4.27	0.0075
Model × Dataset	2	**8.41**	**0.00047**
Family × Dataset	3	1.36	0.260
Model × Family	6	0.60	0.730

TABLE IX
PIXELATION ΔR BY SEVERITY, PASCAL VOC.

Model	sev 1	sev 3	sev 5	Trajectory
YOLOv8n	0.850	0.853	0.864	Flat
YOLOv9c	–	–	0.944	Flat
DETR	0.570	0.573	0.573	Flat
RT-DETR	–	0.648	0.702	Sub-linear

dimension that transfers most strongly across datasets — consistent with both modes reproducing on VOC for essentially every architecture (Sections IV-C.3–IV-C.4). Hallucination, by contrast, is defined in terms of ΔP (Definition 2), the weakest- and most unevenly-transferring dimension, which is consistent with hallucination being only *partially* corruption-intrinsic and sensitive to dataset-specific object statistics. In short: whether a failure mode survives a change of evaluation dataset is, to a substantial degree, predictable from which component of the failure vector defines it.

4) Failure Geometry Transferability Score: Table VII reports FGTS (Section III-H) for the three architectures with complete taxonomy data on both datasets. YOLOv8n has the highest taxonomy stability (FGTS=0.706; 12 of 17 corruptions keep an identical label set on VOC), YOLOv9c is intermediate (FGTS=0.647), and DETR is the least stable (FGTS=0.471; only 8 of 17 corruptions match). Pooled across the three models, 31 of 51 corruption–architecture pairs (=0.608) retain their COCO-derived label on VOC.

FGTS provides a single, model-level number that corroborates, at the level of individual corruptions rather than aggregate mCRS, the same ordering suggested by Table IV: DETR is the architecture whose failure diagnosis is least trustworthy when transferred to a new evaluation benchmark, and the effect is not confined to a single headline corruption (zoom blur) but is visible across roughly a third to a half of the full 17-corruption suite.

5) Formal validation: architecture×dataset interaction: Table VIII reports the two-way ANOVA on severity-5 ΔP described in Section III-I, restricted to YOLOv8n, YOLOv9c, and DETR (the three architectures with complete ΔP data on both datasets). The Model×Dataset interaction is highly significant (F=8.41, p=0.00047), even though RT-DETR — the architecture with the most visually striking sign flip — is entirely excluded from this test. The Dataset main effect is also significant (F=18.12, p=0.00005): ΔP shifts substantially more negative on VOC than on COCO for all three models, consistent with the precision-denominator effect (Section V-C). By contrast, neither the Family×Dataset interaction (p=0.260) nor the Model×Family interaction (p=0.730) reaches significance.

The practical reading of Table VIII is that *how much* and even *which direction* ΔP moves between COCO and VOC depends significantly on which architecture is evaluated, but not on which corruption family is applied. Mean ΔP at severity 5 moves from −0.081 (COCO) to −0.036 (VOC) for DETR — a comparatively small shift — but from −0.082 to −0.275 for YOLOv8n and from −0.061 to −0.307 for YOLOv9c, both substantially larger shifts in the same direction. This establishes that dataset-dependent hallucination / precision behaviour, first flagged as an RT-DETR-specific anomaly in Table V, is in fact a general property of the model×dataset relationship that holds even among the three architectures where RT-DETR's own data quality issue does not apply.

6) Threshold-collapse under pixelation, revisited: Table IX confirms threshold-collapse under pixelation on VOC for three of four architectures: YOLOv8n, YOLOv9c, and DETR all reach >85% of their severity-5 ΔR already at severity 1, mirroring the COCO pattern. RT-DETR shows a milder, sub-linear rise (R^2=0.827 rather than near-flat), but its severity-1 value (0.648, from the severity-3 heatmap read at the earliest available point) is still the majority of its severity-5 value (0.702), so the qualitative signature — most of the damage occurring at low severity — still holds.

7) Late-onset collapse under JPEG compression, revisited: The Late-Onset Collapse mode — mild degradation through severity 3 followed by a sharp acceleration at severity 4–5 — reproduces on VOC for all four models (Table X). The effect is, if anything, more pronounced on VOC than on COCO for YOLOv8n and YOLOv9c, whose severity-5 ΔR (0.750 and 0.760 respectively) is roughly 4–6× their severity-3 values.

This is the strongest cross-dataset confirmation in our study:

TABLE X
JPEG COMPRESSION ΔR: SEVERITY 3 VS. SEVERITY 5, VOC.

Model	sev 3	sev 5	Ratio (sev5/sev3)
YOLOv8n	0.118	0.750	6.4×
YOLOv9c	0.068	0.760	11.2×
DETR	0.163	0.556	3.4×
RT-DETR	0.021	0.201	9.6×

every architecture, on both an 80-category and a 20-category benchmark, shows the same qualitative late-onset signature under JPEG compression, supporting the claim that this failure mode reflects a genuine threshold in JPEG artefact geometry rather than a property of either dataset — and is, as Table VI shows, exactly the kind of ΔR-driven signature our transfer-prediction analysis predicts should replicate most reliably.

8) Failure prediction on VOC: Ridge regression trained on severity 1–3 failure vectors to predict severity 4–5 behaviour achieves an overall R^2 of 0.694 (DETR), 0.832 (YOLOv9c), and 0.738 (RT-DETR) on VOC — comparable to, and in the YOLOv9c case higher than, the COCO figures reported in Section IV.G of the original study. This indicates that the predictive structure of failure geometry is not an artefact of COCO's scale or annotation density. We note that this within-dataset severity-prediction task is conceptually distinct from the cross-dataset failure-vector transfer prediction of Section IV-C3: the former predicts a model's own future (higher-severity) behaviour on one dataset, the latter predicts its behaviour on an entirely different dataset.

D. Dimension Orthogonality on VOC

The ΔR–ΔL correlation on VOC is $r \approx 0.96$–0.99 across models (e.g. 0.99 for DETR, 0.96 for RT-DETR), essentially matching the COCO figures ($r \approx 0.94$–0.96). This confirms that the shared-causal-origin relationship between recall loss and localisation error discussed in Section V.C is not COCO-specific, and is consistent with ΔR and ΔL being precisely the two dimensions shown in Table VI to transfer most reliably across datasets — a correlated pair is, unsurprisingly, also a jointly transferable pair.

V. DISCUSSION

A. Failure as a Structured, Largely Dataset-Independent Phenomenon

The most striking result of this extended study is that most — but not all — of the failure geometry structure identified on COCO survives unchanged on Pascal VOC, and that *which* parts survive is itself predictable. Threshold-collapse under pixelation and late-onset collapse under JPEG compression reproduce with the same qualitative shape on both datasets and across all four architectures, supporting the interpretation that these failure modes are properties of the corruptions themselves — a conclusion now reinforced by the observation that both modes are defined on ΔR, the failure dimension our transfer-prediction analysis shows transfers with $R^2{>}0.88$ for every architecture tested. Hallucination under zoom blur is partially corruption-intrinsic (it survives for three of four models) but is not fully dataset-invariant, and the overall mCRS ranking is stable only for the top-ranked architecture (RT-DETR) — again consistent with hallucination being defined on ΔP, the weakest- and most architecture-dependently transferring dimension we measure.

B. Why Does DETR's Ranking Drop on VOC?

DETR's mCRS falls from a COCO tie for last place to an outright last place on VOC, driven mainly by its noise (0.7668) and blur (0.7532) family scores — the lowest of any model on either dataset. This ranking instability is corroborated at a finer grain by DETR's FGTS of only 0.471 (Section IV-C4) — the lowest of the three architectures we could score — indicating that DETR's individual per-corruption failure-mode diagnoses, not merely its aggregate mCRS, are the least reliable across a change of dataset. Two non-exclusive explanations are consistent with our data. First, DETR's learned object queries and bipartite matching were optimised against COCO's object density and scale distribution; VOC's sparser, larger objects may fall outside the region of query-to-object assignment that DETR handles best under distribution shift, amplifying recall loss under noise and blur. Second, DETR's high operating confidence threshold (0.70, required because DETR emits many low-confidence proposals by default) was tuned for COCO's clean images; under corruption on VOC's different image statistics this fixed threshold may discard a larger share of degraded true-positive detections than the adaptive thresholds implicitly used by the YOLO family. We flag this as a hypothesis supported by, but not conclusively proven by, the present data — testing it directly would require sweeping DETR's threshold on VOC specifically, which we leave to future work.

C. The Precision-Denominator Effect

RT-DETR's ΔP values on VOC are consistently an order of magnitude larger in absolute terms than on COCO or than the other three models on VOC. Because ΔP is normalised by the mean number of ground-truth boxes per image, and VOC images contain fewer annotated objects than COCO images, the same absolute false-positive count change produces a larger ΔP. Critically, our ANOVA (Section IV-C5) shows this is not solely an RT-DETR artefact: the Dataset main effect on ΔP is itself highly significant ($p{=}0.00005$) among the three architectures with complete data, and the Model×Dataset interaction ($p{=}0.00047$) shows the *size* of that shift is architecture-dependent, with YOLOv9c ($-0.061 \rightarrow -0.307$) and YOLOv8n ($-0.082 \rightarrow -0.275$) shifting substantially more than DETR ($-0.081 \rightarrow -0.036$). This is a general lesson for any robustness metric that normalises by object count: cross-dataset comparison of raw ΔP magnitude is not meaningful without either (i) restricting to relative rankings within a dataset, as we do for mCRS, or (ii) renormalising by a dataset-specific reference statistic. We recommend the latter as a direction for future work on cross-dataset failure geometry.

D. Dimension Orthogonality

The correlation between ΔR and ΔL across corruption conditions is $r \approx 0.94$–0.99 for all models on both datasets — substantially higher than desirable for a claimed orthogonal decomposition. This correlation reflects a genuine co-occurrence: corruptions that destroy spatial structure tend to both miss more detections *and* degrade localisation in the remaining ones, driven by the same upstream cause. The dimensions are nevertheless diagnostically distinct: ΔL is defined only over matched true-positive detections, while ΔR captures objects that were not detected at all. We treat this as shared causal origin rather than redundancy, a conclusion strengthened by its replication on VOC and by the fact that these two correlated dimensions are also, as Table VI shows, the two that transfer most reliably across datasets.

E. Deployment Risk

Table XI translates the failure taxonomy into a practitioner-facing risk classification, ranking corruptions by worst-case ΔR at severity 5 across all evaluated models and flagging which risk assessments are confirmed on both COCO and Pascal VOC. Corruptions whose failure mode is ΔR-driven (Threshold- Collapse, Late-Onset Collapse, most Suppression cases) carry a "Critical" or "High" rating that, per Table VI, we have direct quantitative grounds to trust across deployment domains with different object-density statistics. The one "Critical" entry whose underlying mechanism (ΔP-driven hallucination) is explicitly flagged as dataset-dependent is Zoom Blur; practitioners deploying on imagery with markedly different object density than the benchmark used for validation (e.g. moving-camera platforms with sparse scenes) should treat this specific risk assessment with the caveat established in Sections IV-C3 and IV-C5, rather than assuming it transfers with the same reliability as the ΔR-driven risks above it.

TABLE XI
DEPLOYMENT RISK CLASSIFICATION BASED ON WORST-CASE ΔR ACROSS EVALUATED MODELS AT SEVERITY 5, CONFIRMED ON BOTH COCO AND PASCAL VOC.

Corruption	Mode	Risk	Scenario
Pixelate	Threshold-Collapse	Critical	Low-bandwidth transmission
Zoom Blur	Hallucination (dataset-dependent)	Critical	Moving camera platforms
JPEG Sev.5	Late-Onset Collapse	Critical	Compressed video streams
Glass Blur	Suppression	Critical	Dirty / frosted optics
Impulse Noise	Suppression	High	Sensor fault conditions
Motion Blur	Suppression	High	High-speed platforms
Gaussian Noise	Suppression	High	Low-light sensors
Defocus Blur	Gradual	Medium	Fixed-focus cameras
Fog	Gradual	Medium	Outdoor deployment
Snow	Gradual	Medium	Winter environments
Brightness	Gradual	Low	Lighting variation

F. Limitations

Several limitations should inform interpretation of these results.

First, the high per-image standard deviation (mean std ≈ 0.31 on COCO, comparably high on VOC) reflects genuine heterogeneity in both datasets — scenes vary enormously in object count, size distribution, and clutter. Failure vector magnitudes should be interpreted as population-level averages, not per-image predictions.

Second, all four detectors are evaluated on VOC using *COCO-trained* weights, with no VOC-specific fine-tuning. This is intentional — it lets us attribute failure-geometry differences to evaluation-image statistics rather than to differences in training data — but it also means our VOC results characterise cross-domain transfer robustness rather than the robustness of a detector natively trained and deployed on VOC-distributed imagery. A fully controlled study would additionally evaluate VOC-native fine-tuned weights; we leave this to future work.

Third, the ΔP denominator effect described in Section V.C means that raw cross-dataset ΔP magnitudes are not directly comparable; only within-dataset rankings (as used for mCRS) should be treated as robust. We view a dataset-density-normalised ΔP^* metric as a natural and low-cost extension of the present work, and leave its full validation to a subsequent revision.

Fourth, the ΔR–ΔL correlation ($r > 0.94$ on both datasets) indicates that the four dimensions are not fully orthogonal. Future work should explore decoupled formulations of ΔL — for instance, restricting to images where both clean and corrupt predictions yield at least one true positive — to reduce shared variance with ΔR.

Fifth, RT-DETR's COCO evaluation covered 4,952 of 5,000 images (48 skipped due to missing annotations) and detailed per-corruption vectors for several low-degradation weather/digital conditions were not archived for that run; this affects per-family mCRS reliability for RT-DETR on COCO specifically and does not apply to the VOC re-evaluation reported here, which is complete across all 17 corruptions. As a direct consequence, the transfer-prediction (Section IV-C3), FGTS (Section IV-C4), and ANOVA (Section IV-C5) analyses introduced in this extended study are computed on the three architectures with complete bilateral coverage (YOLOv8n, YOLOv9c, DETR); extending them to RT-DETR once its COCO archive is completed is a direct and low-cost item of future work, and would let us test directly whether RT-DETR's zoom-blur sign flip is an extreme instance of the same Model$\times$Dataset interaction we already establish among the other three architectures, or a qualitatively distinct effect.

VI. CONCLUSION

We have introduced Failure Geometry, a framework for decomposing object detection robustness into four orthogonal failure dimensions — ΔR, ΔP, ΔL, ΔE — and, in this extended study, tested it for the first time across two independent detection benchmarks: COCO val2017 and Pascal VOC, four architectures, and 17 corruption types at five severities each (170 evaluation conditions per model in total).

The central empirical finding is that failure behaviour is *largely, but not entirely*, structurally consistent across both architectures and datasets — and, new to this extended study, we show that this consistency is itself measurable and predictable rather than purely qualitative. A cross-dataset transfer-prediction analysis shows that ΔR and ΔL carry over from COCO to VOC with R^2=0.84–0.95, while ΔP and ΔE transfer weakly and architecture-dependently (as low as R^2=0.15 for DETR) — directly explaining why threshold-collapse under pixelation and the newly identified late-onset collapse under JPEG compression (both ΔR-driven) reproduce on VOC for every architecture, while zoom-blur hallucination (defined on ΔP) is only partially corruption- intrinsic and not dataset-invariant in magnitude or sign. The Failure Geometry Transferability Score quantifies this instability directly at the architecture level (YOLOv8n=0.706, YOLOv9c=0.647, DETR=0.471), and a formal two-way ANOVA confirms a statistically significant Model×Dataset interaction on ΔP (F=8.41, p<0.001) even among the three architectures unaffected by RT-DETR's data-completeness limitation — establishing that dataset-dependent hallucination is a general, statistically robust architectural phenomenon rather than a single-model anomaly. The mCRS ranking places RT-DETR highest on both datasets, but the relative ordering of DETR and YOLOv8n reverses between COCO and VOC, a rank instability we trace partly to a precision-normalisation denominator effect driven by VOC's lower average object density.

We hope that the Failure Geometry framework — its five formally defined failure modes, its transfer-prediction and FGTS instruments for *measuring* rather than merely describing cross-dataset consistency, and this first statistically validated cross-dataset stress test of a multi-dimensional robustness taxonomy — provides a richer and more trustworthy vocabulary for robustness evaluation in safety-critical object detection than single-dataset scalar metrics allow.

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